

CMS Open Data H to tau tau Search: Final Analysis Note

Analysis my_analysis

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Contents

Abstract	1
Change Log	1
1 Data And Simulation	1
2 Event Selection And Categories	2
3 QCD And W Control Corrections	3
4 Primary Visible-Mass Result	4
5 Comparison With Published Results	6
6 Conclusion	8
References	9

Abstract

This note documents a reduced CMS 2012 Open Data search for Higgs boson decays to tau pairs in the mu tau_h final state. The final result uses visible mass in VBF, boosted, and zero-jet categories with a same-sign data-driven QCD/fake template. It gives $\mu = 0.4382^{+4.9443}_{-0.4382}$ from a profile-likelihood scan and an observed 95% CLs limit $\mu < 10.7645$.

Change Log

Phase 5 v2 responds to the observed-versus-expected audit. It adds a same-sign QCD/fake estimate, changes the primary full-data result to the visible-mass fallback, removes alternative-method presentation from the final result per the latest user instruction, adds category-level μ diagnostics for the primary method, and compiles the paper with REVTeX PRL formatting.

1 Data And Simulation

The data are Run2012B and Run2012C TauPlusX reduced samples from the public HiggsTauTauReduced mirror, using the CMS Open Data H to tau tau luminosity reference of 11.467/fb (CMS Collaboration 2021; CMS Open Data Analyses 2021). The localized mirror files contain about one tenth of the event entries listed in the Open Data records; this is treated as the public reduced processing sample scope, while MC weights continue to use the official record denominators specified in `normalization_inputs.json`, as required by the sample provenance.

The available MC set contains ggH and VBF H to tau tau, DYJetsToLL, TTbar, and W1/W2/W3JetsToLNu. QCD multijet, diboson, single top, embedded Z to tau tau, W4/inclusive W+jets, associated Higgs, and H to WW remain unavailable as reduced samples and are not silently substituted.

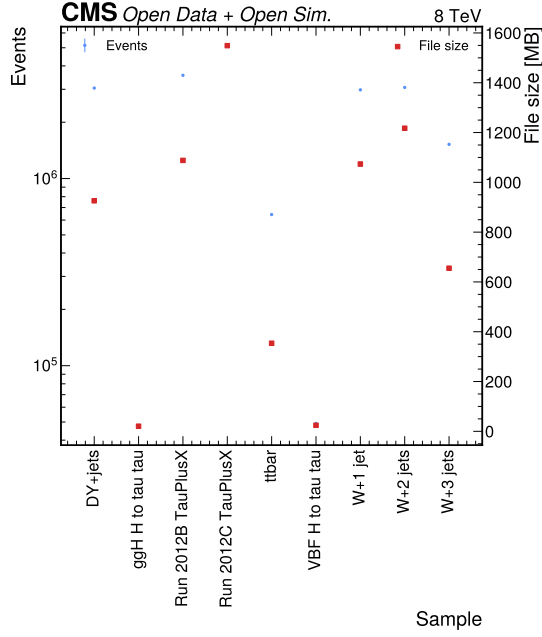


Figure 1: Sample inventory. The figure summarizes local reduced ROOT entries and file sizes for the public H to tau tau files used in the analysis. The entries are processing entries, while normalization uses official Open Data record denominators and the cited luminosity reference.

2 Event Selection And Categories

Events are required to pass `HLT_IsoMu17_eta2p1_LooseIsoPFTau20`. Muons have $pt > 20$ GeV and $|\eta| < 2.1$ with tight identification and isolation. Hadronic tau candidates have $pt > 20$ GeV, $|\eta| < 2.3$, decay-mode ID, tight isolation ID, and tight anti-muon ID. The signal region uses opposite-sign mu tau_h pairs with muon-MET transverse mass below 40 GeV. The W control region uses high transverse mass, and the QCD/fake estimate uses same-sign low-mT events.

The fit categories are mutually exclusive VBF, boosted, and zero-jet. VBF requires at least two clean jets, leading-dijet mass above 300 GeV, and leading-dijet $|\Delta\eta|$ above 2.5. Boosted events have at least one clean jet after failing VBF, and zero-jet events have no clean jets.

Category	Data	Background	QCD/fake	Signal	Data/background
vbf	78	96.28	17.75	1.590	0.810
boosted	2183	2364.39	325.58	9.170	0.923
zero_jet	8451	8456.81	1569.65	14.341	0.999

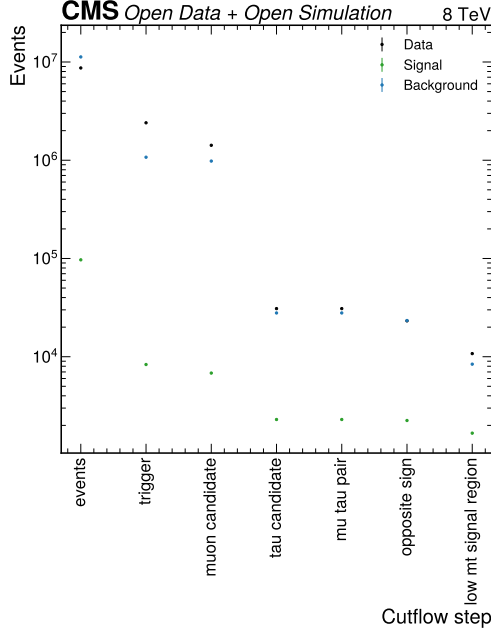


Figure 2: Cutflow summary. This plot shows raw event counts after each selection step for data, signal MC, and background MC. It verifies that the selection chain is monotonic and that the reduced samples remain populated after trigger, object, charge, and transverse-mass requirements.

3 QCD And W Control Corrections

The W+jets normalization is measured in the high-mT control region as 0.8528 ± 0.0370 . The 10% validation value was 0.8433 ± 0.0868 . The values agree within the larger 10% uncertainty and support the W-control arithmetic.

The VBF category originally had a large MC overprediction. Tightening the VBF cuts and applying a simple b-tag veto did not solve it, so a disjoint VBF-like top-btag control region outside the low-mT signal region was used to calibrate the MC-background rate in the VBF category. The resulting VBF background scale is 0.5571 ± 0.0515 ; it is applied only to MC backgrounds in VBF and not to the signal or QCD/fake template.

The same-sign QCD/fake estimate subtracts non-QCD MC from same-sign low-mT data and transfers the resulting template to opposite-sign signal candidates. For the primary visible-mass model, the OS/SS transfer factor measured in the lowest visible-mass bin is 0.9958 ± 0.1201 .

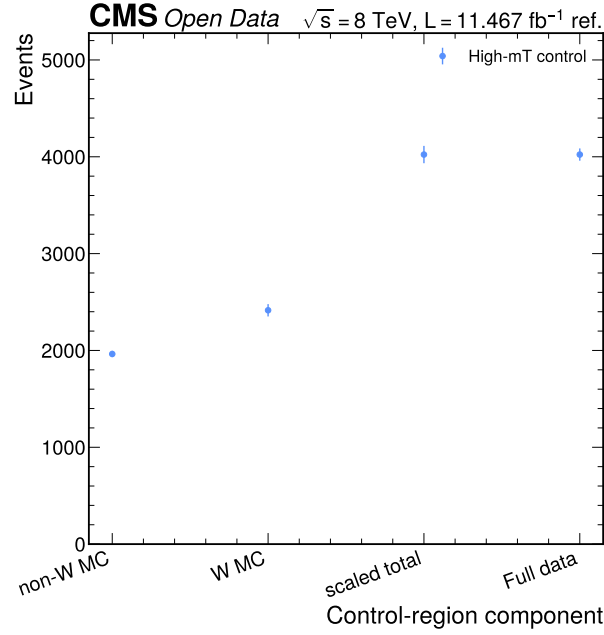


Figure 3: Full high-mT W control comparison. The figure shows the derivation inputs for the full-data W+jets control scale. The control region is outside the low-mT signal region and the scale is propagated to the observed workspace without signal-region tuning.

4 Primary Visible-Mass Result

The primary statistical model is a binned pyhf profile likelihood in visible mass, with one signal-strength parameter μ , the W high-mT scale, the same-sign QCD/fake transfer uncertainty, luminosity, DY and tau/open-data normalization terms, and MC statistical terms (Heinrich et al. 2021; Read 2002; Cowan et al. 2011).

Quantity	Value	Interpretation
Primary median expected 95% CLs limit	$\mu < 10.7375$	corrected visible-mass workspace
Primary observed 95% CLs limit	$\mu < 10.7645$	conservative observed result
Primary μ central value	$0.4382 +4.9443 -0.4382$	profile likelihood $q(\mu)=1$ interval
Primary q_0 Z	0.0949	diagnostic only
Primary combined data/background	0.9812	validation after QCD/fake correction
Primary χ^2/ndf	0.7224	validation after QCD/fake correction

The primary result does not show a Higgs-like excess: μ is $0.4382 +4.9443 -0.4382$ and the observed 95% CLs limit is $\mu < 10.7645$. The broad interval is compatible with the weak sensitivity expected from this reduced single-final-state setup.

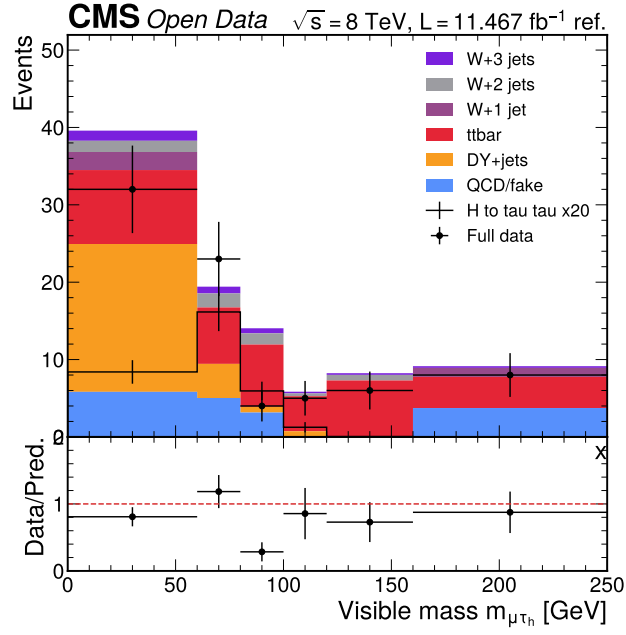


Figure 4: Primary visible-mass validation in VBF. The plot compares full data to the QCD-corrected visible-mass prediction in the VBF category. The remaining VBF deficit is a limitation, but it no longer drives a fake signal-strength increase.

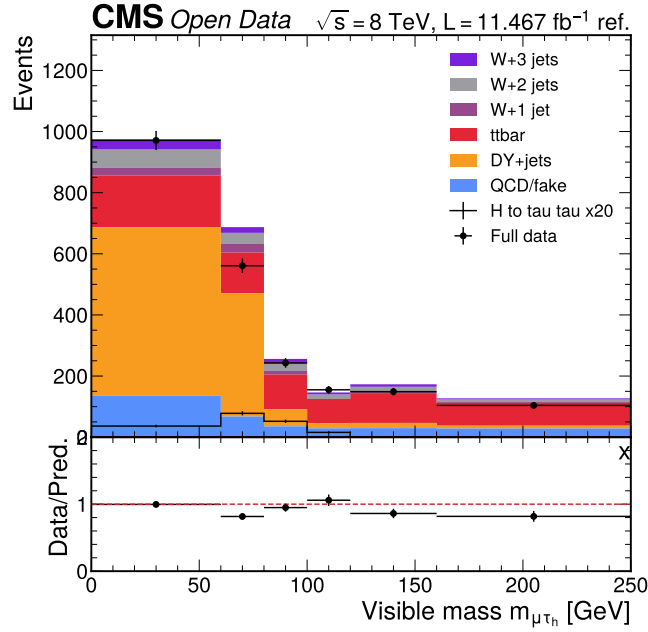


Figure 5: Primary visible-mass validation in boosted. The plot compares full data to the QCD-corrected visible-mass prediction in the boosted category. This channel is part of the conservative primary fit.

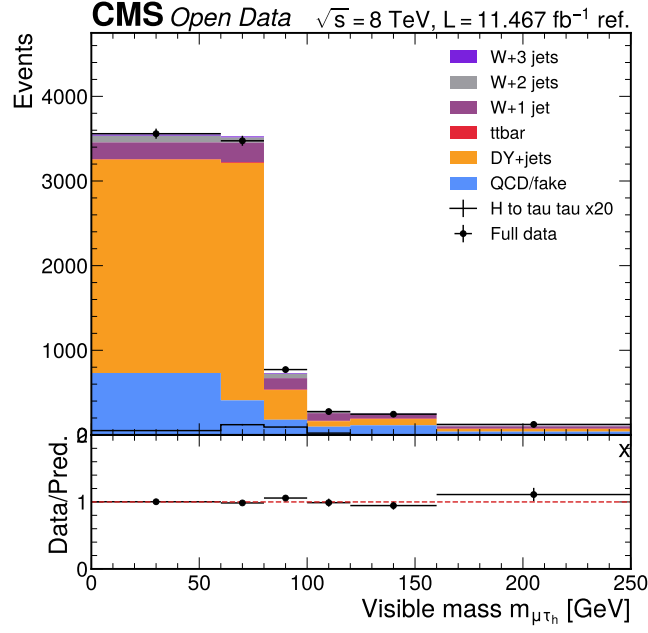


Figure 6: Primary visible-mass validation in zero-jet. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate, which removes the dominant normalization pathology seen before the QCD/fake correction.

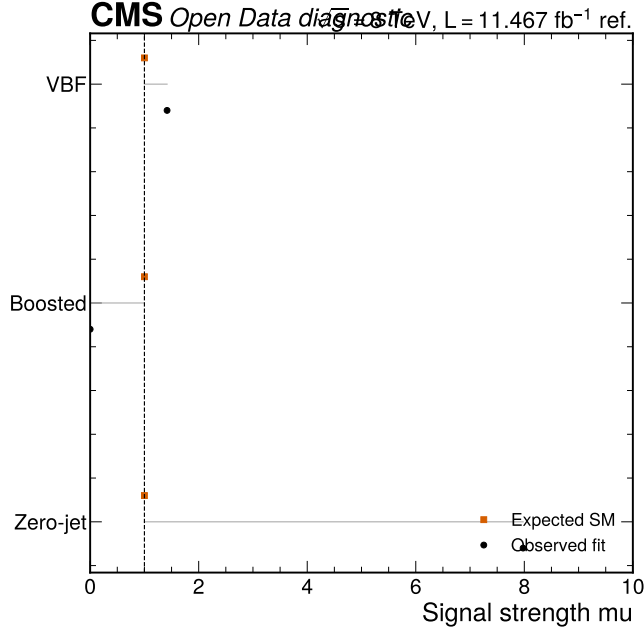


Figure 7: Category signal-strength comparison. The figure shows the Standard Model expectation $\mu = 1$ and the observed single-category profile-fit value for VBF, boosted, and zero-jet categories using the primary visible-mass model. These category fits are diagnostics; the quoted result remains the simultaneous three-category fit.

5 Comparison With Published Results

CMS Run 1 reported evidence for H to tau tau using 7 and 8 TeV data with multiple final states, embedded backgrounds, and a full calibration program; the quoted best-fit signal strength was 0.78 ± 0.27 with observed and expected significances of 3.2 and 3.7 standard deviations (CMS Collaboration 2014). CMS 2018 reported observed and expected significances of 4.9 and 4.7 in the 2016 data, and 5.9 when combined with earlier CMS data, with a signal strength near 1.09 (CMS Collaboration 2018). The ATLAS+CMS Run 1 combination gives broader global

Higgs-rate context, while the PDG/Higgs-summary SM H to tau tau branching fraction provides the denominator convention for $\mu = 1$ rather than a single-channel open-data comparison target (ATLAS Collaboration and CMS Collaboration 2016; Navas et al. 2024; Particle Data Group 2024; LHC Higgs Cross Section Working Group 2017).

Result	Scope	Significance	Signal-strength information
This analysis	8 TeV μ tau_h reduced mirror, visible mass	0.095	$\mu = 0.438 +4.944 -0.438$; 95% CLs $\mu < 10.764$; expected $\mu < 10.738$
CMS Run 1 JHEP 2014	7+8 TeV multi-channel	observed 3.2, expected 3.7	$\mu = 0.78 \pm 0.27$
CMS PLB 2018 2016 data	13 TeV multi-channel	observed 4.9, expected 4.7	$\mu = 1.09 \pm \text{about } 0.27$
CMS PLB 2018 combined	CMS 7+8+13 TeV combination	observed 5.9, expected 5.9	same signal-strength model as CMS 2018 publication
ATLAS+CMS Run 1 combination	7+8 TeV global Higgs couplings/rates	H to tau tau evidence context	global $\mu = 1.09 \pm 0.11$
PDG / HXSWG SM context	H to tau tau branching fraction at mH near 125 GeV	not a search significance	BR(H to tau tau) about 6.3%, used only as SM normalization context

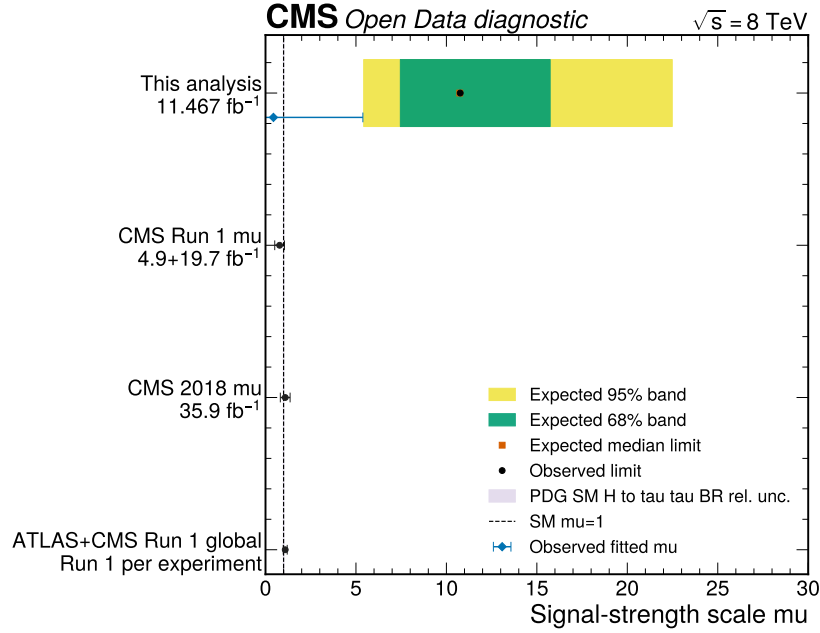


Figure 8: Signal-strength and limit comparison. The figure uses a CMS-style expected-band and observed-marker presentation for the primary open-data limit, published CMS/ATLAS+CMS signal-strength measurements, and a narrow PDG SM H to tau tau branching-ratio normalization band at $\mu = 1$. The open-data diagnostic is not directly equivalent to CMS publication measurements.

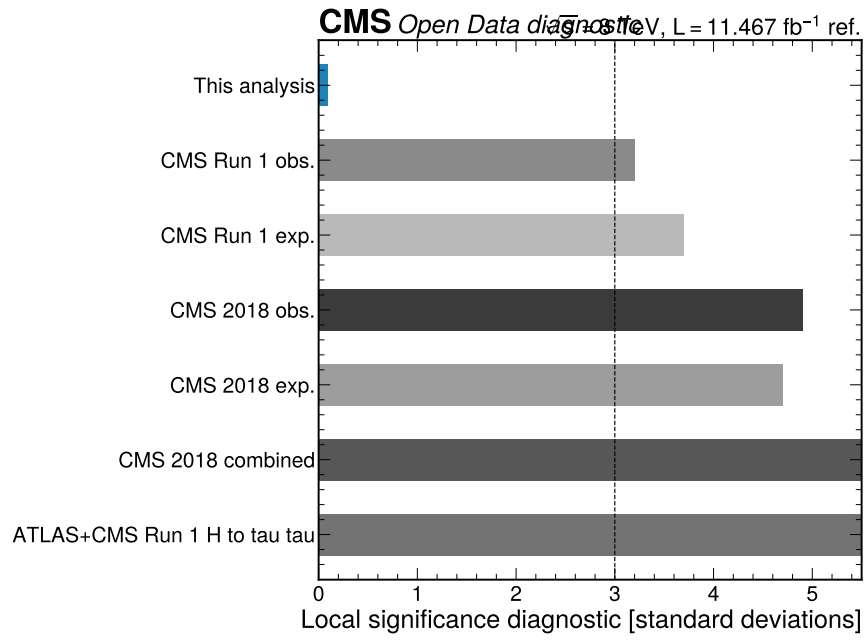


Figure 9: Significance comparison. The figure compares the primary open-data diagnostic with CMS and ATLAS+CMS publication values. Only the primary open-data result is shown from this analysis in the final result figure.

6 Conclusion

The final audit-corrected result is a conservative visible-mass plus QCD/fake template fit. It gives $\mu = 0.4382 +4.9443 -0.4382$ and an observed 95% CLs limit $\mu < 10.7645$. This is the correct interpretation of the reduced CMS Open Data workflow: reproducible and useful for methodology, but not CMS-quality evidence for H to tau tau.

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